

JEE MAIN REVISION MASTER: DIFFERENTIAL EQUATIONS 2026 3 QUESTIONS 5% WEIGHTAGE

↑50.15%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 27 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Differential Equations - Core Application	27 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Differential Equations - Core Application (Very High Yield — 27 PYQs)

Core Principle: Key problem pattern in Kcet Differential Equations.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Sum of the squares of the order and degree (if defined) of a differential equation $2y''+(y')^2=\sqrt{(y')^2-3}$ is..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If $\int xf(x)dx+(f(x)/2)=0$, then $f(x)$ is equal to..." [Ans: C]
- [2026 Shift] Q#3 (MCQ): "Integrating factor of the differential equation $(1+x^2)(dy/dx)+xy=1$ is..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Sum of the squares of the order and degree (if defined) of a differential equation $2y'' + (y'')^2 = \sqrt{y''} - 3$ is

- (A) 3
- (B) 20 **[Correct]**
- (C) 8
- (D) 16

Correct Answer: B

Solution Analysis: The given differential equation is $2y'' + (y'')^2 = \sqrt{y''} - 3$. The highest order derivative present is y'' , so the order of the differential equation is 2. To find the degree, the equation must be written as a polynomial in derivatives. Since y'' is inside a square root, we first remove the radical by squaring both sides: $(2y'' + (y'')^2)^2 = (y'' - 3)$. Expanding, $4(y'')^2 + 4y''(y'')^2 + (y'')^4 = y'' - 3$. Bring all terms to one side: $(y'')^4 + 4y''(y'')^2 + 4(y'')^2 - y'' + 3 = 0$. Now it is a polynomial in derivatives. The highest order deriva